R16/R15/R13

Code No: 135CD/125AM/115AM

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, March - 2021
ELECTRONIC MEASUREMENTS AND INSTRUMENTATION
(R16-Common to ECE, CSE; R15-Electronics and Communication Engineering; R13-Electronics and Communication Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) What are the Dynamic characteristics of measuring systems? Define them.
b) Explain the working of Average responding Voltmeters. [9+6]

2.a) Describe the different types of errors.
b) How could you extend the range of Voltmeter? Explain with an example. [7+8]

3.a) What is the need of spectrum analyzers?
b) Discuss about pulse and square wave generator. [6+9]

4.a) Describe the working of AF Signal generator.
b) Draw the block diagram Heterodyne wave analyzer and explain its working. [6+9]

5.a) How do you measure Period and Frequency using CRO? Explain.
b) Compare Dual Beam and Dual trace CROs. [6+9]

6.a) What are the elements CRT and describe the function of each element of it.
b) Draw the Block diagram of Digital storage CRO and explain function of each block. [7+8]

7.a) Explain the working principle of Synchros and mention its applications.
b) Describe the principles of operation of capacitive transducers and mention their applications. [7+8]

8.a) How do you measure humidity? Explain.
b) Draw the block diagram of Data Acquisition System and explain the function of each block. [7+8]

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